

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EMT 315 — Past Questions & Materials
by Renance

Course: EMT 315
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 300

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Paper 1 of 2 — EMT 315 Statistics for Environmental science

Level 300 • First Semester • PastQuestion

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Koshineke Emmanuel
cos/9419/2022

DEPARTMENT OF ENVIRONMENTAL MANAGEMENT AND TOXICOLOGY
COLLEGE OF SCIENCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES

FIRST SEMESTER 2024/2025 UNDERGRADUATE EXAMINATION

EMT 315: Statistics for Environmental sciences (2 units)

Time: 2:30 hours

INSTRUCTION: Answer four (4) questions in all, (NB: Two Question from each section)

SECTION A

1. A university administrator wants to analyze the academic performance of postgraduate students in a statistics course. The scores (out of 100) of a randomly selected sample of 15 students are given below (Scores: 78, 85, 92, 88, 74, 69, 95, 80, 83, 77, 90, 72, 88, 91, 84) Using descriptive statistics, summarize the dataset by computing the following:
 - a. Measures of Central Tendency: (i) Mean (ii) Median (iii) Mode
 - b. Measures of Dispersion: (i) Range (ii) Variance (iii) Standard Deviation
 - c. Comment briefly on the distribution of scores based on the computed statistics.
2. A government agency is conducting a nationwide survey to assess public perception of environmental policies. You are responsible for designing the sampling strategy.
 - a. Discuss four different sampling techniques that could be used.
 - b. Which technique would be most appropriate for this survey and why?
3. A hospital wants to evaluate the effectiveness of a new diet plan for diabetic patients.
 - a. Explain the difference between primary and secondary data in this study.
 - b. What type of data collection method would you recommend and why?

mc 71.5

SECTION B

4. An environmental scientist studying the impact of temperature on plants growth collects data on these variable. The data is as follows

Sn	Temp(Degree Celsius)	Plant growth (cm)
1	20	10
2	22	12
3	25	15
4	28	18
5	30	20

Assuming the data is normally distributed

- a) What statistic would be appropriate to determine the relationship between temperature and plant growth
 - b) Using the appropriate statistics, find the correlation coefficient.
 - c) Based on the correlation coefficient what could be said about the relationship between temperature and plant growth.
5. The ministry of environmental claims that the average PH level of rivers in the state is 7.2. A researcher collects water samples from 10 different locations and measures the PH levels. The result is as follows: 7.1, 7.8, 6.9, 7.0, 7.4, 6.9, 7.2, 7.5, 7.8 and 6.9. Using an α level of 0.05, is there enough evidence to support the claim that the average PH level in the state is more than 7.2
6. (a) Explain the following; (i) Parametric test (ii) Non parametric test
 (b) List three examples each of parametric and Non-parametric test clearly stating scenario where they can be applied
 (c) List the steps involved in testing an hypothesis.

Emmanuel
cos/94/9/2022

t Table

cum. prob one-tail two-tails	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
df	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
Z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
Confidence Level											

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Paper 2 of 2 — EMT 315 STATISTICS FOR ENVIRONMENTAL SCIENCES 2024I2025

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mc715

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